INCH-POUND
MS27654D
w/AMENDMENT 2
27 June 2025
SUPERSEDING
MS27654D
w/AMENDMENT 1
8 February 2008

DETAIL SPECIFICATION SHEET

CONNECTORS, RECEPTACLE, ELECTRICAL, BACK PANEL, WALL MOUNTING FLANGE, CRIMP TYPE, BAYONET COUPLING, SERIES I

Inactive for new design after 20 February 1978. For new design, use MS27656.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification and MIL-DTL-38999.

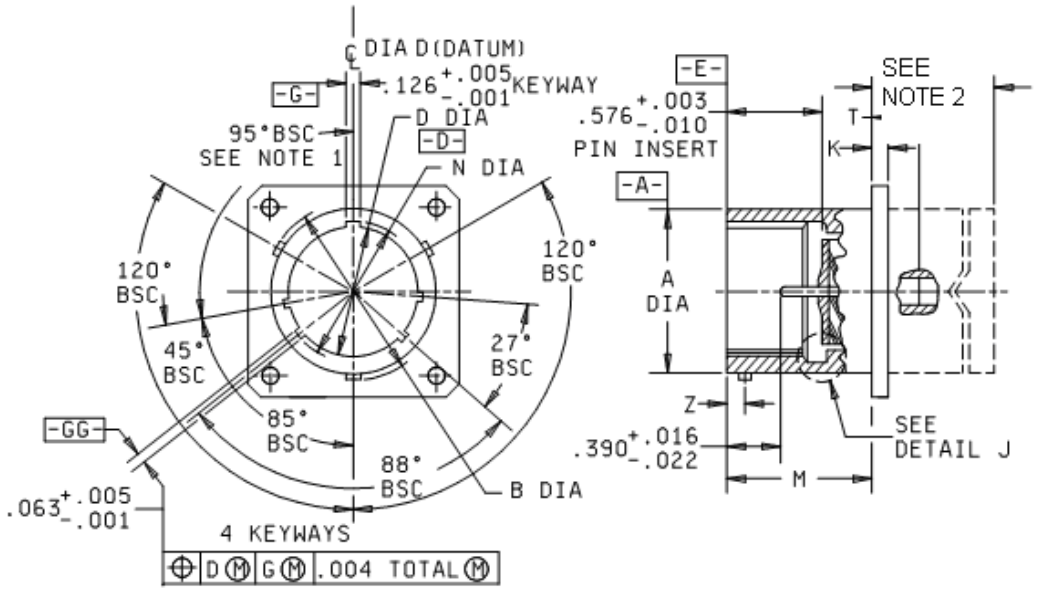
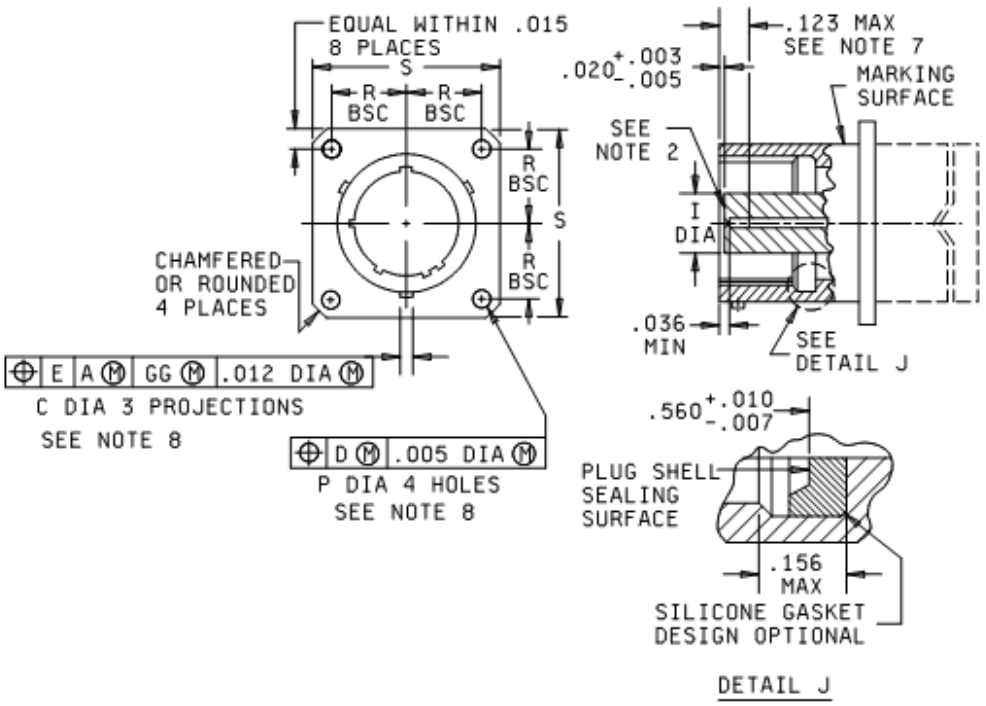


FIGURE 1. Receptacle, classes E and T.

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Shell size	A dia +.001 -.005	B dia +.000, -.016 over projection	C dia +.006 -.002	D dia +.005 -.001	I dia max	K +.015 -.000	M +.000 -.005
9	.572	.662	.078	.442	.290	.085	.820
11	.700	.810		.570	.418		
13	.850	.960		.687	.532		
15	.975	1.085		.812	.657		
17	1.100	1.210		.937	.782		
19	1.207	1.317		1.042	.871		
21	1.332	1.442		1.167	.996	.115	.790
23	1.457	1.557		1.292	1.121		
25	1.582	1.692	.125	1.417	1.246		

FIGURE 1. Receptacle, classes E and T – Continued.

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Shell size	N dia +.005 -.006	P dia +.010 -.005	R BSC	S ± .020	T max	Z BSC	For insert arrangement see MIL-STD-1560
9	.492	.128	.3595	.938	.156	.100	
11	.620		.406	1.031			
13	.769		.453	1.125			
15	.894		.4845	1.219			
17	1.019		.531	1.312			
19	1.124		.578	1.438			
21	1.249		.625	1.562	.186		
23	1.374	.147	.6875	1.688			
25	1.499		.750	1.812		.109	

FIGURE 1. Receptacle, classes E and T - Continued.

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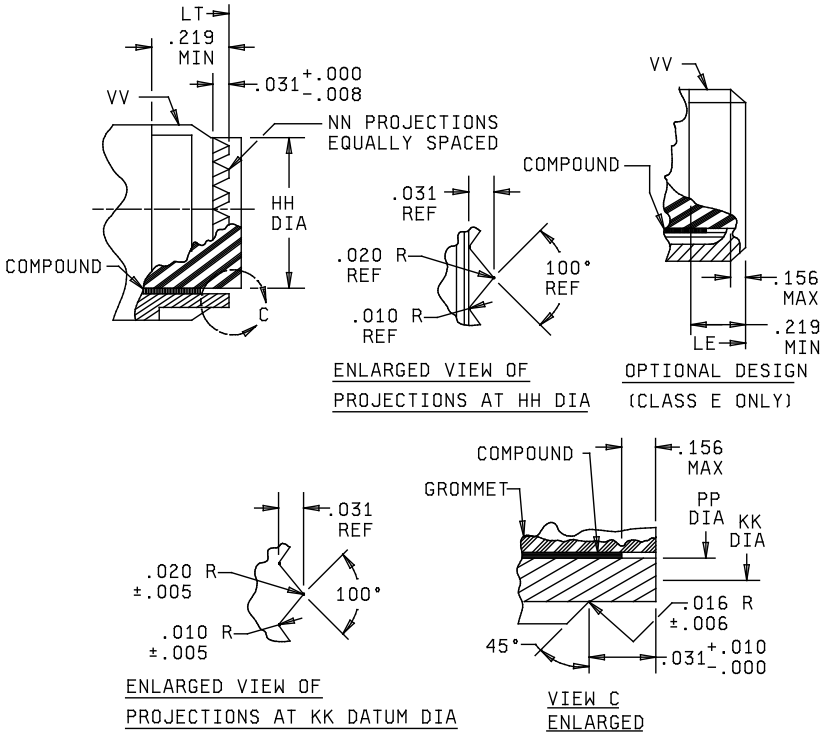


FIGURE 1. Receptacle, classes E and T – Continued.

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Inches	mm	Inches	mm	Inches	mm	Inches	mm	Inches	mm
.000	0.00	.128	3.25	.570	14.48	.850	21.59	1.207	30.66
.001	0.03	.147	3.73	.572	14.53	.853	21.67	1.210	30.73
.002	0.05	.156	3.96	.576	14.63	.871	22.12	1.218	30.94
.003	0.08	.188	4.78	.578	14.68	.879	22.33	1.219	30.96
.004	0.10	.219	5.56	.586	14.88	.894	22.71	1.235	31.37
.005	0.13	.290	7.37	.609	15.47	.937	23.80	1.246	31.65
.007	0.18	.344	8.74	.620	15.75	.938	23.83	1.249	31.72
.008	0.20	.3595	9.13	.625	15.88	.942	23.93	1.292	32.82
.010	0.25	.366	9.30	.628	15.95	.960	24.38	1.312	33.32
.012	0.30	.387	9.83	.657	16.69	.974	24.74	1.317	33.45
.015	0.38	.390	9.91	.662	16.81	.975	24.77	1.332	33.83
.016	0.41	.406	10.31	.687	17.45	.985	25.02	1.340	34.04
.020	0.51	.418	10.62	.6875	17.46	.996	25.30	1.360	34.54
.022	0.56	.442	11.23	.700	17.78	1.019	25.88	1.374	34.90
.031	0.79	.453	11.51	.711	18.06	1.031	26.19	1.417	35.99
.036	0.91	.469	11.91	.734	18.64	1.042	26.47	1.438	36.53
.063	1.60	.472	11.99	.750	19.05	1.067	27.10	1.442	36.63
.078	1.98	.4845	12.31	.754	19.15	1.085	27.56	1.457	37.01
.085	2.16	.487	12.37	.769	19.53	1.097	27.86	1.499	38.07
.100	2.54	.492	12.50	.782	19.86	1.100	27.94	1.562	38.67
.109	2.77	.500	12.70	.790	20.07	1.110	28.19	1.567	39.80
.115	2.92	.515	13.08	.810	20.57	1.121	28.47	1.582	40.18
.123	3.12	.531	13.49	.812	20.62	1.124	28.55	1.688	42.88
.125	3.18	.532	13.51	.820	20.83	1.167	29.64	1.692	42.98
.126	3.20	.560	14.22	.836	21.23	1.192	30.28	1.812	46.02

FIGURE 1. Receptacle, classes E and T – Continued.

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Shell size	HH dia +.000 -.005	KK DATUM dia	NN	PP dia ± .003	LE max overall length	LT max overall length	VV Thread class 2A UNEF (plated)
9	.387	.366	12	.344	.469	.469	7/16-28
11	.515	.487	16	.472			9/16-24
13	.628	.609	20	.586			11/16-24
15	.754	.731	24	.711			13/16-20
17	.879	.853	28	.836			15/16-20
19	.985	.974	32	.942			1 1/16-18
21	1.110	1.097	36	1.067	.500	.500	1 3/16-18
23	1.235	1.218	40	1.192			1 5/16-18
25	1.360	1.340	44	1.317			1 7/16-18

NOTES:

1. Normal keyway position. For other keyway positions, see MIL-DTL-38999.
2. Insert front surface shall be flat within .005 Total Indicator Reading (TIR).
3. Diameters A and D shall be concentric within .015 TIR.
4. Diameters D and I shall be concentric within .015 TIR at Maximum Material Condition (MMC).
5. Diameter A with respect to diameter B, and diameter D with respect to diameter N shall be concentric at MMC.
6. Dimensions are in inches. Metric equivalents are given for general information only.
7. The point at which the gage pin, having the same basic diameter as the mating contact and a square face, touches socket contact spring.
8. The gage features for GG shall be .010 smaller than the MMC size at basic location.

FIGURE 1. Receptacle, classes E and T - Continued.

REQUIREMENTS:

Dimensions and configuration: See figure 1.
Interface dimensions shall conform to MIL-DTL-38999.
Mating connectors: MS27467, MS27653 and MS27661.
Insert arrangements: See MIL-STD-1560.

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Part or Identifying Number (PIN) example:



Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Referenced documents. In addition to MIL-DTL-38999, this document references the following:

MS27467
MS27653
MS27656
MS27661
MIL-STD-1560

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - AS
Air Force - 11
DLA - CC

Preparing activity:
DLA - CC

(Project 5935-2025-056)

Review activities:
Army - AR, MI
Navy - EC, MC, OS
Air Force - 19

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